Hugo Ruivo

Senior Software Engineer | CGI Pipelines & Production Systems

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Professional Summary

Senior software engineer and Real-Time R&D Lead with 10+ years across automotive CGI, real-time 3D, and simulation. Designs connected production systems spanning vehicle data, artist-facing tools, Python services, Unreal Engine, and rendering infrastructure. Leads collaborative engineering work from architecture through delivery. Personal projects explore compilers, telemetry, RISC-V, and FPGA design.

Core Skills

Professional languages: Python, C++, TypeScript. Personal systems work: Zig, RISC-V, compiler development, VHDL/FPGA.

Web and services: SvelteKit, Bun, FastAPI, REST APIs; Docker deployment for web frontends and backends.

Production engineering: shared data structures, Unreal Engine tools, Perforce stream architecture, custom Deadline plugins, Jenkins CI/CD, Git, Houdini, Blender.

Engineering and delivery: C++/Unreal integration, simulation, profiling, automation tests, technical leadership with developers and artists.

Professional Experience

Real-Time R&D Lead | Digital Giant

May 2022 - Present

- Design a multi-brand automotive CGI pipeline that normalizes brand-specific imports into common vehicle data for shared Unreal Engine tools.
- Design and integrate SvelteKit/Bun frontends and Python/FastAPI REST services; deploy the web components with Docker.
- Develop production tooling, Unreal Engine rigging tools, and custom Deadline plugins for render submission from Unreal or a browser.
- Design Perforce stream organization for shared vehicle data, reusable plugins, and project workflows.
- Build Jenkins CI/CD with Python, shell tooling, and Unreal Engine automation tests.

Case study: hruivo.github.io/work/automotive-cgi-pipeline/

CGI Real-Time R&D Lead | Hogarth Worldwide

2020 - 2022

- Led development of an Unreal Engine vehicle configurator with multiple environments and cinematic sequences.
- Integrated web-based systems with Unreal Engine for product validation and stakeholder feedback.

Technical Artist | Burrows

2019 - 2020

- Created Unreal Engine tools for vehicle import, configuration, and real-time rendering workflows.
- Built a WebGL vehicle configurator with an automated Houdini pipeline to optimize complex assets for browser rendering.

Software Developer | Mackevision

2018 - 2019

- Maintained Unreal Engine vehicle configurators using Pixel Streaming and extended interactive functionality.
- Implemented communication between an iPad control interface and the main Unreal Engine application.

Software Developer | Abyssal (now Ocean Infinity)

2016 - 2018

- Developed an Unreal Engine remotely operated vehicle simulator for the oil and gas industry.
- Implemented buoyancy, tether physics, hydraulic manipulator simulation, multi-display rendering, and custom hardware input integration.
- Designed modular vehicle systems, data-driven missions, and session recording; led developers and artists as the prototype expanded.

Game Programmer | X-Team

2012 - 2016

- Developed web and mobile game projects using Unity.

Selected Personal Projects

DCS Autonomous Aircraft

Flight firmware using Lunara for autonomous takeoff, waypoint navigation, and landing of the F-16 and Su-25 in DCS. Telion streams telemetry to Mission Dashboard and relays commands to the aircraft.

Lunara Language and Compiler

Zig compiler targeting bare-metal RISC-V (RV32I), inspired by NASA/JPL's Power of Ten rules. Includes unit-aware types, checked arithmetic, mandatory wait timeouts, fault handling, mission scenarios, and fault injection.

Telemetry and Mission Control

Telion telemetry backend and a web dashboard for monitoring KSP/DCS vehicles and sending commands. Built an adapter to display NASA Artemis II telemetry using the same data format and dashboard.

CHIP-8 FPGA Core

CHIP-8 implemented in VHDL, running on the GameBub console's FPGA.

AI Assisted Car Configurator

Integrated AI agents with an Unreal Engine vehicle configurator using Python, C++, and OpenAI/llama models.

Education

Game Development Engineering | Polytechnic Institute of Cavado and Ave

Barcelos, Portugal | 2009 - 2012